Introduction to Radiobiological Models

Souvik Das

Resisdent Medical Physicist
Tata Memorial Hospital, Mumbai

Table of Contents

1. Linear Quadratic (LQ) Model
2. Nominal Standard Dose (NSD)
3. Cumulative Radiation Effect (CRE)
4. Time-Dose-Fractionation (TDF)

Table of Contents

1. Linear Quadratic (LQ) Model
2. Nominal Standard Dose (NSD)
3. Cumulative Radiation Effect (CRE)
4. Time-Dose-Fractionation (TDF)

Target theory

- The target of radiation for the purpose of therapy is the **DNA**.

Target theory

- The target of radiation for the purpose of therapy is the **DNA**.
- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.

Target theory

- The target of radiation for the purpose of therapy is the **DNA**.
- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.
- **Poisson statistics** are used for the basis for survival equations.

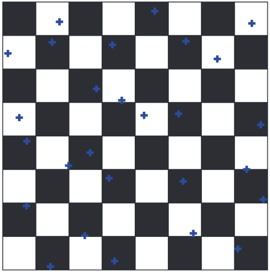


Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

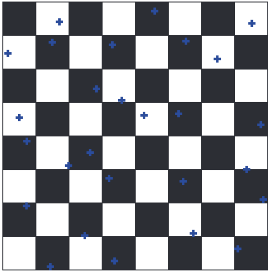


Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

- Poisson statistics describe a large number of random events happening to a large number of subjects, averaging out to a small number of events per subject.

- If x is the average number of drops per square, then the probability of a square being hit by k drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

- If x is the average number of drops per square, then the probability of a square being hit by k drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

- The probability of a square being dry is given by:

$$P(0) = e^{-x}$$

- If x is the average number of drops per square, then the probability of a square being hit by k drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

- The probability of a square being dry is given by:

$$P(0) = e^{-x}$$

- The probability of a square being wet is given by:

$$P(\text{wet}) = 1 - e^{-x}$$

- The Poisson model is used for surviving fraction (**SF**) of cells:

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:

- The Poisson model is used for surviving fraction (**SF**) of cells:
    - At an average of x lethal hits per cell:
        - e^{-x} cells survive (no hits).

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,

Poisson Distribution
Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. $\text{SF} = 0.37$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. $\mathrm{SF} = 0.37$
    - 2 hits kills 86% of cells. $\mathrm{SF} = 0.14$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. SF $= 0.37$
    - 2 hits kills 86% of cells. SF $= 0.14$
    - 3 hits kills 95% of cells. SF $= 0.05$

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. $\text{SF} = 0.37$
    - 2 hits kills 86% of cells. $\text{SF} = 0.14$
    - 3 hits kills 95% of cells. $\text{SF} = 0.05$
    - 4 hits kills 98% of cells. $\text{SF} = 0.02$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. SF $= 0.37$
    - 2 hits kills 86% of cells. SF $= 0.14$
    - 3 hits kills 95% of cells. SF $= 0.05$
    - 4 hits kills 98% of cells. SF $= 0.02$
    - 5 hits kills 99% of cells. SF $= 0.01$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. SF $= 0.37$
    - 2 hits kills 86% of cells. SF $= 0.14$
    - 3 hits kills 95% of cells. SF $= 0.05$
    - 4 hits kills 98% of cells. SF $= 0.02$
    - 5 hits kills 99% of cells. SF $= 0.01$

Note

D_0 is defined as the radiation dose resulting in 37% survival. This is assumed to be equal to the radiation dose necessary to cause one **lethal** hit per cell.

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 tumor cell per pt.: $\text{TCP} = 0.37$
    - 2 tumor cell per pt.: $\text{TCP} = 0.14$
    - 3 tumor cells per pt.: $\text{TCP} = 0.05$
    - 4 tumor cells per pt.: $\text{TCP} = 0.02$
    - 5 tumor cells per pt.: $\text{TCP} = 0.01$

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 tumor cell per pt.: TCP $= 0.37$
    - 2 tumor cell per pt.: TCP $= 0.14$
    - 3 tumor cells per pt.: TCP $= 0.05$
    - 4 tumor cells per pt.: TCP $= 0.02$
    - 5 tumor cells per pt.: TCP $= 0.01$

Note

To achieve a certain TCP, you should aim for a tumor cell survival of (1 - TCP).

Target theory

Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.
- The survival fraction (SF) is given by:

$$SF = e^{-D/D_0}$$

where D is the dose and D_0 is the dose required to reduce the surviving fraction to 37%.

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.
- The survival fraction (SF) is given by:

$$SF = e^{-D/D_0}$$

  where D is the dose and D_0 is the dose required to reduce the surviving fraction to 37%.
- This type of survival curve is seen for mammalian cells which are irradiated with **high LET** radiation such as alpha particles or carbon ions; very **radiation-sensitive** cells like lymphocytes and bone marrow cells; cells that have major defects in DNA doublestrand break repair; are synchronized in M phase; and for cells that have DNA double-strand break repair inhibited by chemical or gene knockdown.

Target theory

Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.

Target theory

Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.
- For a single dose D, the survival fraction (SF) is given by:

$$SF = 1 - (1 - e^{-D/D_0})^n$$

where D is the dose, D_0 is the dose required to reduce the surviving fraction to 37%, and n is the number of targets.

Target theory

Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.
- For a single dose D, the survival fraction (SF) is given by:

$$SF = 1 - (1 - e^{-D/D_0})^n$$

  where D is the dose, D_0 is the dose required to reduce the surviving fraction to 37%, and n is the number of targets.
- This type of survival curve is seen for mammalian cells which are irradiated with **low LET** radiation such as X-rays or gamma rays; cells that are **radiation-resistant** like fibroblasts and epithelial cells; cells that have a normal DNA double-strand break repair; and for cells that are synchronized in G1 or S phase.

Target theory

Multitarget, Single-Hit Model

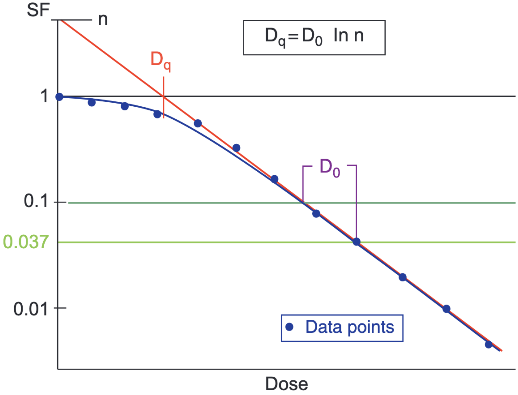


Figure: The single-hit, multitarget survival curve. Notice that it is curved at low doses and straight (e.g., survival is an exponential function of dose) at high doses. This allows you to calculate D_0 by taking measurements in the high-dose region

Target theory
Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at $SF = 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between $SF = 0.1$ and $SF = 0.037$.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at $SF = 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between $SF = 0.1$ and $SF = 0.037$.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF $= 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF $= 0.1$ and SF $= 0.037$.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
  - $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from SF $= 0.1$ to SF $= 0.01$).

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF $= 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF $= 0.1$ and SF $= 0.037$.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
  - $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from SF $= 0.1$ to SF $= 0.01$).
- The low-dose portion of the curve is known as the "shoulder":

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF $= 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF $= 0.1$ and SF $= 0.037$.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
  - $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from SF $= 0.1$ to SF $= 0.01$).
- The low-dose portion of the curve is known as the "shoulder":
  - D_q tells you how wide the shoulder is.

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF $= 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF $= 0.1$ and SF $= 0.037$.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
  - $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from SF $= 0.1$ to SF $= 0.01$).
- The low-dose portion of the curve is known as the "shoulder":
  - D_q tells you how wide the shoulder is.
  - D_q is a measure of the cell's repair capacity. More repair means a larger D_q and a larger shoulder

- Advantages:

- Advantages:
  - The high-dose portion of the curve is a straight line.

Single-Hit, Multitarget Model

Advantages and Disadvantages

- Advantages:
    - The high-dose portion of the curve is a straight line.
    - This straight-line component correlates well with cell-culture experiments.

- Advantages:
  - The high-dose portion of the curve is a straight line.
  - This straight-line component correlates well with cell-culture experiments.
- Disadvantages:

- Advantages:
  - The high-dose portion of the curve is a straight line.
  - This straight-line component correlates well with cell-culture experiments.
- Disadvantages:
  - The low-dose portion of the curve greatly underestimates cell kill.

Single-Hit, Multitarget Model
Advantages and Disadvantages

- Advantages:
    - The high-dose portion of the curve is a straight line.
    - This straight-line component correlates well with cell-culture experiments.
- Disadvantages:
    - The low-dose portion of the curve greatly underestimates cell kill.
    - Unlike the linear-quadratic model, the single-hit model is not based on a molecular mechanism.

Single-Hit, Multitarget Model

Advantages and Disadvantages

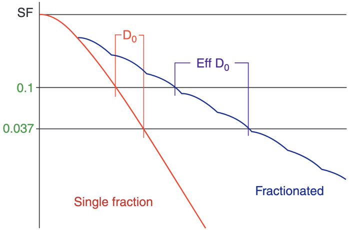
- Advantages:
  - The high-dose portion of the curve is a straight line.
  - This straight-line component correlates well with cell-culture experiments.
- Disadvantages:
  - The low-dose portion of the curve greatly underestimates cell kill.
  - Unlike the linear-quadratic model, the single-hit model is not based on a molecular mechanism.
  - Unlike the linear-quadratic model, there is not a simple equation for "equivalent dose" given a daily fractionation schedule.

Fractionated Radiation and Effective D_0

- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.

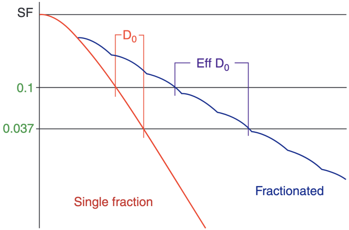
Fractionated Radiation and Effective D_0

- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.
- Assuming complete repair in between fractions, the shoulder is repeated with each fraction.

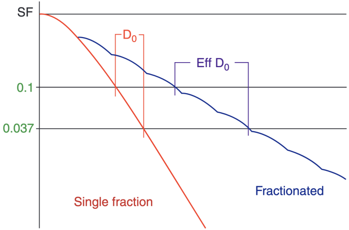


Fractionated Radiation and Effective D_0

- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.
- Assuming complete repair in between fractions, the shoulder is repeated with each fraction.
- D_0 versus effective D_0. When radiation is fractionated, you need much more dose to achieve the same amount of cell kill. Therefore, effective D_0 is always larger than D_0



- The slope of a fractionated survival curve is called effective D_0.



Fractionated Radiation and Effective D_0

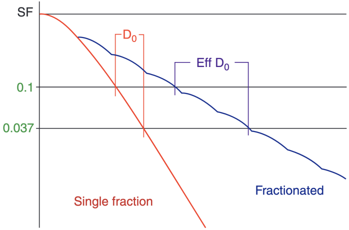
Continue

- The slope of a fractionated survival curve is called effective D_0.
- For a single fraction of D Gy with a surviving fraction of SF_D:

$$SF_D = e^{-D/D_0}$$

$$SF_D = e^{-D/\text{effective } D_0}$$

$$\text{effective } D_0 = -\frac{D}{\ln(SF_D)}$$



Fractionated Radiation and Effective D_0

Continue

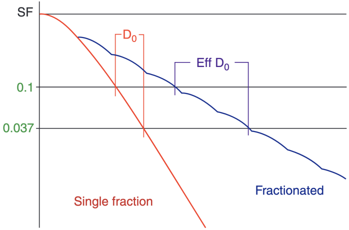
- The slope of a fractionated survival curve is called effective D_0.
- For a single fraction of D Gy with a surviving fraction of SF_D:

$$\mathrm{SF}_D = e^{-D/D_0}$$

$$\mathrm{SF}_D = e^{-D/\text{effective } D_0}$$

$$\text{effective } D_0 = -\frac{D}{\ln(\mathrm{SF}_D)}$$

- After x fractions to total dose xD Gy: $SF_x^D = e^{-xD/\text{effective } D_0}$



Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D.

Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D.
- Lethal DNA aberrations $= \alpha D + \beta D^2 =$ Cell Kill

$$\boxed{SF_D = e^{-(\alpha D + \beta D^2)}}$$

Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D.
- Lethal DNA aberrations $= \alpha D + \beta D^2 =$ Cell Kill

$$\boxed{SF_D = e^{-(\alpha D + \beta D^2)}}$$

- Unlike the single-hit model, the LQ model accounts for two different types of lethal hits which are based on known molecular mechanisms of DNA damage:

Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D.
- Lethal DNA aberrations $= \alpha D + \beta D^2 =$ Cell Kill

$$\boxed{SF_D = e^{-(\alpha D + \beta D^2)}}$$

- Unlike the single-hit model, the LQ model accounts for two different types of lethal hits which are based on known molecular mechanisms of DNA damage:
  - Single-hit kill (α) is unrepairable damage and is **independent** of fractionation or dose rate. This corresponds to single-hit and **intra**-track accumulated damage.

Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D.
- Lethal DNA aberrations $= \alpha D + \beta D^2 =$ Cell Kill

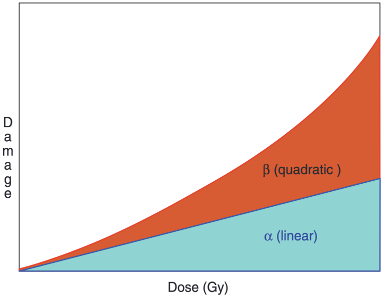
$$\boxed{SF_D = e^{-(\alpha D + \beta D^2)}}$$

- Unlike the single-hit model, the LQ model accounts for two different types of lethal hits which are based on known molecular mechanisms of DNA damage:
  - Single-hit kill (α) is unrepairable damage and is **independent** of fractionation or dose rate. This corresponds to single-hit and **intra**-track accumulated damage.
  - Multi-hit kill (β) is repairable damage and is **dependent** on fractionation or dose rate. This corresponds to double-hit and **inter**-track accumulated damage.

Linear-Quadratic (LQ, Alpha-Beta) Model

Continue:

- The α/β ratio is the **dose** at which α kill and β kill are equal.
  - Low α/β ratio ("high repair") tissues are relatively resistant at small fraction size and relatively sensitive at large fraction size.
  - High α/β ratio ("low repair") tissues are relatively sensitive at small fraction size and relatively resistant at large fraction size.



- LQ model is widely used in radiotherapy to predict the effects of different fractionation schemes on both tumor and normal tissues.

Linear-Quadratic (LQ, Alpha-Beta) Model

Alpha-Beta Model and Dose Fractionation

- LQ model is widely used in radiotherapy to predict the effects of different fractionation schemes on both tumor and normal tissues.
- Suppose we have n no of fractions of dose d each, then the total dose $D = nd$. The surviving fraction after n fractions is given by:

$$
\begin{aligned}
\mathrm{SF}_n &= e^{-n(\alpha d + \beta d^2)} \\
\implies -\frac{1}{nd} &= \frac{\alpha}{\ln(\mathrm{SF}_n)} + \frac{\beta d}{ln(\mathrm{SF}_n)}
\end{aligned}
$$

- LQ model is widely used in radiotherapy to predict the effects of different fractionation schemes on both tumor and normal tissues.
- Suppose we have n no of fractions of dose d each, then the total dose $D = nd$. The surviving fraction after n fractions is given by:

$$\mathrm{SF}_n = e^{-n(\alpha d + \beta d^2)}$$

$$\implies -\frac{1}{nd} = \frac{\alpha}{\ln(\mathrm{SF}_n)} + \frac{\beta d}{ln(\mathrm{SF}_n)}$$

- Plot of dose per fraction d versus $-\frac{1}{nd}$ gives a straight line with slope $\beta/ln(\mathrm{SF}_n)$ and y-intercept $ln(\mathrm{SF}_n)$. The ratio of intercept to slope gives α/β.

Linear-Quadratic (LQ, Alpha-Beta) Model

- LQ model is widely used in radiotherapy to predict the effects of different fractionation schemes on both tumor and normal tissues.
- Suppose we have n no of fractions of dose d each, then the total dose $D = nd$. The surviving fraction after n fractions is given by:

$$\mathrm{SF}_n = e^{-n(\alpha d + \beta d^2)}$$

$$\implies -\frac{1}{nd} = \frac{\alpha}{\ln(\mathrm{SF}_n)} + \frac{\beta d}{ln(\mathrm{SF}_n)}$$

- Plot of dose per fraction d versus $-\frac{1}{nd}$ gives a straight line with slope $\beta/ln(\mathrm{SF}_n)$ and y-intercept $ln(\mathrm{SF}_n)$. The ratio of intercept to slope gives α/β.
- Such values are obtained for a variety of tissues and tumors, and are used to guide clinical decision-making regarding fractionation schemes.

- Most acute-reacting tissues are believed to have an α/β ratio $\approx$ 10.
- Most late-reacting tissues are believed to have an α/β ratio $\approx$ 3 or less.
- CNS tissue (brain, cord) is believed to have an α/β ratio $\approx$ 1-2.
- Most tumors are believed to have an α/β ratio $\geq$ 10.
- Some low α/β tumors may have an α/β ratio $\approx$ 1.5-4.Breast and Prostate are the "classic" low α/β tumors.

Linear-Quadratic (LQ, Alpha-Beta) Model

Biologically Effective Dose

- One advantage of the linear-quadratic model is that it is easy to calculate effective doses for different fraction sizes.

Definition

Biologically effective dose (BED) is an extrapolated dose given over infinitely many fractions.

- For n fractions of d dose per fraction:

$$\begin{aligned}\text{BED}_{\alpha/\beta} &= nd\left(1 + \frac{d}{\alpha/\beta}\right)\\ &= D\left(1 + \frac{d}{\alpha/\beta}\right)\end{aligned}$$

- BED $=$ total physical dose $\times$ relative effectiveness of that dose.

Explain the consequences of unplanned gaps on tumour cells during radiotherapy treatment.

- **Repopulation** of tumour cells may occur.
- The unplanned gap during radiotherapy will cause the cells within a **tumour to proliferate** and leads to an increase in the number of cells that the radiation needs to kill.
- There is evidence that **repopulation can be accelerated in the treated tumours** so that this effect can be significant.
- Thus an increase of overall treatment time (due to the gap) will decrease the efficacy of radiotherapy.

Treatment Interruption

Recommendations

Give the recommendations about how to compensate for unscheduled gaps in radiotherapy treatment.

Treatment Interruption

Recommendations

Give the recommendations about how to compensate for unscheduled gaps in radiotherapy treatment.

1. The main recommendation is to compensate for a gap by **continuing to deliver the same number of fractions in the same overall time**, by inserting the missed fractions (using the original dose per fraction) either during weekends or as extra fraction on previously treatment days.

Give the recommendations about how to compensate for unscheduled gaps in radiotherapy treatment.

1. The main recommendation is to compensate for a gap by **continuing to deliver the same number of fractions in the same overall time**, by inserting the missed fractions (using the original dose per fraction) either during weekends or as extra fraction on previously treatment days.
2. When multiple fractions are given in a day, the interval between them should be **at least 6 hours**.

Treatment Interruption

Recommendations

Give the recommendations about how to compensate for unscheduled gaps in radiotherapy treatment.

1. The main recommendation is to compensate for a gap by **continuing to deliver the same number of fractions in the same overall time**, by inserting the missed fractions (using the original dose per fraction) either during weekends or as extra fraction on previously treatment days.
2. When multiple fractions are given in a day, the interval between them should be **at least 6 hours**.
3. If the break in treatment occurs early in the schedule, then two fractions per day can be used. However, if there are not enough days left on the original schedule to catch up using two fractions on each remaining days, three fractions per day can be considered subjected to the availability of resources.

MCQ Example

Question 1: Which of the following remains unaltered while changing the plan from fractionated radiotherapy to brachytherapy?

1. Total Dose
2. Dose per fraction
3. BED
4. Relative effetiveness of that dose

MCQ Example

Question 1: Which of the following remains unaltered while changing the plan from fractionated radiotherapy to brachytherapy?

3. **BED**

MCQ Example

Question 2: A tumour containing 10^8 cells is inactivated by radiation with a mean lethal dose of $D_0 = 1.8$ Gy. What is the number of surviving tumour cells after a radiation dose of 45 Gy?

1. 1.2×10^{-3}
2. 1.4×10^{-4}
3. 1.4×10^{-3}
4. 1.2×10^{-4}

MCQ Example

Question 2: A tumour containing 10^8 cells is inactivated by radiation with a mean lethal dose of $D_0 = 1.8$ Gy. What is the number of surviving tumour cells after a radiation dose of 45 Gy?

3. 1.4×10^{-3}

Solution

The number of surviving cells is given by:

$$N = N_0 e^{-D/D_0}$$

where N_0 is the initial number of cells, D is the dose, and D_0 is the mean lethal dose. Substituting the values, we get:

$$N = 10^8 e^{-45/1.8} \approx 1.4 \times 10^{-3}$$

Problem 1

If a treatment is planned for 70 Gy/ 35 fractions, but due to departmental contraints, we need to complete it in 25 fractions. What is the new dose per fraction for same tumor control with $\alpha/\beta = 10$?(Ans: 2.65 Gy)

- EQD_2 is the dose in 2 Gy fractions that would produce the same biological effect as the actual treatment.
- For n fractions of d dose per fraction:

$$\text{EQD}_2 = \frac{BED}{1 + \frac{2}{\alpha/\beta}}$$
$$\implies \text{EQD}_2 = D\left(\frac{d + \alpha/\beta}{2 + \alpha/\beta}\right)$$

Problem 2

You want to treat a lung cancer to 60 Gy in 2 Gy fractions, but the first five fractions are given at 3 Gy/fx due to SVC syndrome. Assuming α/β $= 3$, how much additional dose should you deliver at 2 Gy per fraction? (Ans: 42 Gy)

Problem 3

A 40-year-old lady with colorectal cancer is being treated with radio therapy. A dose of 45 Gy in 25 daily fractions over 5 weeks is planned. After ten fractions, treatment was cancelled for five fractions due to machine breakdown. It was planned to deliver the isoeffective tumour dose by increasing the size of the ten fractions to finish the treatment as scheduled. By using the "equivalent dose" method, calculate required dose per fraction for the last ten fractions. Assume $\alpha/\beta = 10$ Gy for colorectal cancer. (Ans: 2.54 Gy)

Table of Contents

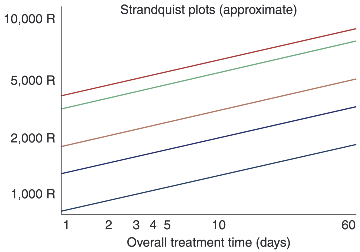
1. Linear Quadratic (LQ) Model
2. Nominal Standard Dose (NSD)
3. Cumulative Radiation Effect (CRE)
4. Time-Dose-Fractionation (TDF)

Need for time-dose models

- The aim of the radiotherist is to obtain high probability of tumor control with minimal normal tissue complications.
- Since typical treatment plans are quite complicated to compare so with the help of this models. the treatment regimes can be easily modified.
- Time dose relationship have been evolved in different forms like graphical, tabular and mathematical forms. The most commonly used models are:
    - Nominal Standard Dose (NSD)
    - Cumulative Radiation Effect (CRE)
    - Time Dose Fractionation (TDF)

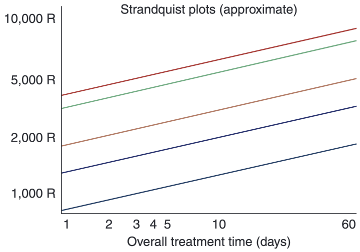
Nominal Standard Dose (NSD)

- An empiric equation based on the Strandquist Plots:



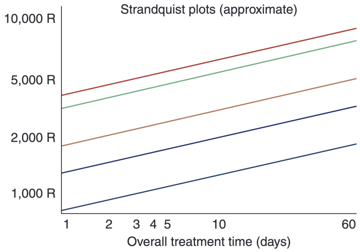
Nominal Standard Dose (NSD)

- An empiric equation based on the Strandquist Plots:
  - Back in the 1930s-1940s, Strandquist treated a bunch of skin cancers and plotted radium exposure versus skin erythema, desquamation, necrosis, and tumor cure.



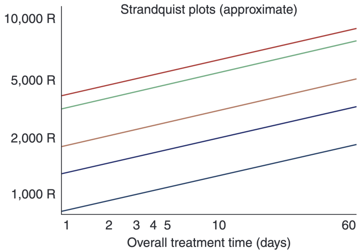
Nominal Standard Dose (NSD)

- An empiric equation based on the Strandquist Plots:
    - Back in the 1930s-1940s, Strandquist treated a bunch of skin cancers and plotted radium exposure versus skin erythema, desquamation, necrosis, and tumor cure.
    - He found that the relationship between total dose and overall treatment time formed a series of parallel lines on a logarithmic chart.



Nominal Standard Dose (NSD)

- An empiric equation based on the Strandquist Plots:
    - Back in the 1930s-1940s, Strandquist treated a bunch of skin cancers and plotted radium exposure versus skin erythema, desquamation, necrosis, and tumor cure.
    - He found that the relationship between total dose and overall treatment time formed a series of parallel lines on a logarithmic chart.
    - The Strandquist plots (approximate). These parallel lines represent various skin-related endpoints such as erythema, desquamation, and necrosis.



Nominal Standard Dose (NSD)

Continue

- Unlike newer models, the Ellis NSD has no theoretical basis whatsoever; it is an empiric **"curve fitting"** model.

Nominal Standard Dose (NSD)
Continue

- Unlike newer models, the Ellis NSD has no theoretical basis whatsoever; it is an empiric **“curve fitting”** model.
- The NSD can accurately predict acute skin toxicity and skin cancer response because that is where the data came from.

Nominal Standard Dose (NSD)

Continue

- Unlike newer models, the Ellis NSD has no theoretical basis whatsoever; it is an empiric **“curve fitting”** model.
- The NSD can accurately predict acute skin toxicity and skin cancer response because that is where the data came from.
- The NSD makes no attempt to predict late toxicity.

Nominal Standard Dose (NSD)

Continue

- Unlike newer models, the Ellis NSD has no theoretical basis whatsoever; it is an empiric **“curve fitting”** model.
- The NSD can accurately predict acute skin toxicity and skin cancer response because that is where the data came from.
- The NSD makes no attempt to predict late toxicity.
- There are two versions of the equation: one with time and fractionation, another with just fractionation.

Nominal Standard Dose (NSD)

Continue

- Unlike newer models, the Ellis NSD has no theoretical basis whatsoever; it is an empiric **"curve fitting"** model.
- The NSD can accurately predict acute skin toxicity and skin cancer response because that is where the data came from.
- The NSD makes no attempt to predict late toxicity.
- There are two versions of the equation: one with time and fractionation, another with just fractionation.
- For N fractions of total dose D(in rads) delivered over T days:

$$D = \text{NSD} \times N^{0.24} \times T^{0.11}$$

Table of Contents

1. Linear Quadratic (LQ) Model
2. Nominal Standard Dose (NSD)
3. Cumulative Radiation Effect (CRE)
4. Time-Dose-Fractionation (TDF)

Cumulative Radiation Effect (CRE)

- NSD formula does not say anything about tumor responce and normal tissue tolerace due to part of the treatment schedule which had already been completed, might have reached some level of tolerance.

Cumulative Radiation Effect (CRE)

- NSD formula does not say anything about tumor responce and normal tissue tolerace due to part of the treatment schedule which had already been completed, might have reached some level of tolerance.
- Kirk et. al. extended the NSD concept to the cumulative radiation effect (CRE) for such type of schedules. This is based on sub-tolerance of normal tissue which was subjected to irradiation.

Cumulative Radiation Effect (CRE)

- NSD formula does not say anything about tumor responce and normal tissue tolerace due to part of the treatment schedule which had already been completed, might have reached some level of tolerance.
- Kirk et. al. extended the NSD concept to the cumulative radiation effect (CRE) for such type of schedules. This is based on sub-tolerance of normal tissue which was subjected to irradiation.
- The partial tolerances are additive in nature,

$$PT = \sum PT_i$$

Table of Contents

1. Linear Quadratic (LQ) Model
2. Nominal Standard Dose (NSD)
3. Cumulative Radiation Effect (CRE)
4. Time-Dose-Fractionation (TDF)

Time-Dose-Fractionation (TDF)

- Orton and Ellis(1973) developed time dose and fractionation concept to make use of partial tolerance more practicable. The TDF for fractionated beam therapy was given by:

$$\text{TDF} = d^{1.538} \times n \times x^{-0.169} \times 10^{-3}$$

where d = dose per fraction, n = number of fractions, x = t/n, t = overall treatment time in days.